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Prediction of thermal contact resistance for reusable heat-pipe cooled thermal protection system based on an inverse thermo-mechanical coupling method

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ABSTRACT

Thermal contact resistance (TCR) is a critical characteristic on increasing or decreasing thermal energy transmission efficiency between two bodies in thermal management systems. However, it is difficult to accurately determine the thermal contact resistance in actual engineering structures, due to the complicated influence factors, such as pressure, temperature, and physical properties. In the work, a new method is presented to estimate the thermal contact resistance for a three-dimensional (3D) reusable heat-pipe cooled thermal protection system based on boundary measurements, by solving transient inverse thermo-mechanical coupling problems. Moreover, the thermal contact resistance varies with interface pressure, temperature, and spatial position, which is more practical and challenging. The thermo-mechanical coupling analysis is conducted by the finite element method, and the inversion is carried out by the gradient-based algorithm. First, the present method is validated by identifying the constant TCR, based on the available experimental data. Then, the thermal contact resistance with the functional form in the 3D reusable thermal protection system is accurately estimated. Finally, the convergence stability and robustness of the proposed method are evaluated, by considering the effects of initial guess value and measurement error, respectively. The accurate determination of thermal contact resistance of the reusable thermal protection system effectively avoids the overweight of aircraft and improves its utilization. The present work provides a novel approach for the determination of thermal contact resistance in actual engineering applications.

Nomenclature

c_p	= specific heat capacity, J/(kg·K)
$\boldsymbol{D}$	= constitutive matrix
F	= objective function
h	= convective heat coefficient, W/(m^2·K)
$\boldsymbol{J}$	= sensitivity matrix
k	= thermal conductivity, W/(m·K)
M	= total number of measurements
P	= interface pressure, Mpa
Q	= volumetric heat source, W/m^3
q	= heat flux, W/m^2
q_r	= radiative heat flux, W/m^2
R	=thermal contact resistance, m^2·K/W
T	= temperature, K

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t	=time, s
u, v, w	= displacements in x, y, and z directions, m
$\boldsymbol{X}$	= inverted parameters
x, y, z	= coordinates, m
Greek	
$\boldsymbol{\theta}$	= updated vector of recovered/inverted parameters
ξ	= random measurement error
η	= normal distributed random variable
ε	= strain
ε_r	= emissivity
σ	= normal stress, Pa
σ_b	= Stefan-Boltzmann constant, 5.67×10^{-8} W/(m^2·K^4)
τ	= shear stress, Pa
ρ	= density, kg/m^3

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(*continued*)

α	= thermal expansion coefficient, 1/K
ζ	= small positive number
μ	=damping factor
ω	= relaxation factor
Subscripts	
∞	= ambient
0	= initial time
Superscripts	
*	= measured value
p	= iteration number

1. Introduction

Thermal contact resistance (TCR) significantly influences the heat transfer process in many thermal management systems, such as thermal transport in flexible electronic devices [1], contact behaviors between the heat exchanger and the storage material in thermal energy storage (TES) system [2], and thermal protection system (TPS) in hypersonic vehicles [3–5]. Fig. 1 shows a reusable heat-pipe cooled thermal protection system that ensures the spacecraft performs multiple missions [6], in which the thermal contact resistance is between the wall and heat pipe. In the design of spacecraft thermal protection systems, underestimation of thermal contact resistance will make spacecraft heavy and inefficient, while overestimation of thermal contact resistance will affect the safety of spacecraft. Therefore, it is very essential to accurately determine thermal contact resistance in practical engineering.

Numerous methods [7–9] have been proposed to determine the thermal contact resistance. Experiment studies [10–12] are considered as accurate approaches, but the results of invasive and non-invasive measurements are different. Besides, it is hard to measure the thermal contact resistance with complex structures and thin-walled materials, and there are few experiments to measure the TCR at high temperature. Numerical methods [13,14] require reconstruction of rough surfaces, and only constant thermal contact resistance can be obtained in each prediction. However, it is well known that thermal contact resistance of multiple component systems is affected by interface pressure, temperature, surface roughness, physical properties, etc. complicated factors [15,16]. Hence, direct numerical simulations are not suitable for complex engineering applications.

Under this circumstance, the inverse method [17–21] is proposed to predict thermal contact resistance or its reciprocal thermal contact conductance/interface conductance based on easily-obtained measurement data, which does not require reconstruction of rough surfaces and is convenient to be used in engineering problems. Guo et al. [5] estimated time-dependent thermal contact conductance for one-dimensional (1D) multilayered insulation composite materials, based on experimental data and the conjugate gradient method. The positions of thermal sensors required in the inverse heat conduction problem (IHCP) are in the interior of the material. Parikh et al. [22] predicted thermal contact conductance for four different one-dimensional heat conduction problems used the Jaya algorithm, and sensor location as well as measurement error etc. influence factors were taken into consideration. Goudarzi et al. [23] used artificial neural networks (ANN) to identify the thermal contact conductance in the exhaust valve of internal combustion engine, and the thermocouples are located in the interior of the calculated region. It concluded that the Levenberg-Marquardt (LM) algorithm provides the best training performance in the model. In addition to directly estimating the thermal contact resistance, some indirect inversion methods also were used. Kanjanakijkasem [24] estimated spatially varying thermal contact resistance of constant, sinusoidal, and triangle functions based on six interior measurement points by solving boundary inverse heat conduction problems, and the constant form of TCR can be accurately captured. Fieberg and Kneer [25] predicted the contact heat flux by solving IHCP, and then the thermal contact resistances of a steel-steel and a steel-aluminum alloy combinations were calculated. Liu et al. [26] determined the thermal contact conductance by identifying the interfacial heat flux based on interior measurements, and studied heat transfer behaviors between Al–Mg–Si alloy and die material at different contact conditions.

Fig. 1. Reusable heat-pipe cooled thermal protection system [6].

As mentioned above, simple one-dimensional models were widely used in the identification of thermal contact resistance. Nevertheless, thermal contact resistance varies with materials, geometries, and working conditions, and the 1D models cannot simulate actual engineering applications very well. Moreover, most researchers determined thermal contact resistance by solving inverse heat transfer problems. The thermo-mechanical coupling inversion of TCR, which takes into account the effects of contact pressure, interface temperature, spatial position, etc. factors, is rarely carried out, to the best of the authors' knowledge. In addition, the invasive measurement method modifies the thermal behavior of the bodies and the measurements may be inaccurate, especially for the thin-walled structures. Thus, it is significant to identify the thermal contact resistance based on boundary measurement information in inversion. The non-invasive measurement method is convenient to be operated.

The heat-pipe cooled thermal protection system effectively transfers heat from the stagnation area to flat surface and radiates it away externally to space, which is a reusable system to reduce the peak temperature and thermal stress of materials. In the present work, we estimate the thermal contact resistance based on the boundary measurement information, by solving transient inverse thermo-mechanical coupling problems of a three-dimensional (3D) reusable heat-pipe cooled thermal protection system. The thermal contact resistance with the functional form includes the effects of pressure, temperature, and spatial position, which is more general and challenging. The finite element method in ABAQUS software is applied to solve the thermo-mechanical coupling problem, which has the advantage of dealing well with nonlinear problems in engineering applications. The gradient-based Levenberg-Marquardt algorithm [27–29] does not need global search, the results stable and accurate, therefore, it is used to carry out the inversion.

The remainder of the work is as follows. In Section 2, theoretical modeling including direct and inverse problems for the 3D reusable heat-pipe cooled thermal protection system is described. Section 3 is the validation of the identification method based on available experimental data. The temperature and stress distributions of the reusable heat-pipe cooled thermal protection system are described in Section 4. The accuracy and robustness of the present method for estimating TCR are analyzed in Section 5, and the conclusions are drawn in Section 6.

2. Theoretical modeling

2.1. Description of the direct problem

The three-dimensional reusable heat-pipe cooled thermal protection system consists of two components with different materials as shown in Fig. 2, and the thermal contact resistance is between the wall and heat pipe.

For the thermal equilibrium condition, the governing equation of three-dimensional transient heat transfer problems can be defined as follows:

$$\frac{\partial}{\partial x}\left[k(T)\frac{\partial T}{\partial x}\right]+\frac{\partial}{\partial y}\left[k(T)\frac{\partial T}{\partial y}\right]+\frac{\partial}{\partial z}\left[k(T)\frac{\partial T}{\partial z}\right]+Q=\rho(T)c_p(T)\frac{\partial T}{\partial t} \tag{1}$$

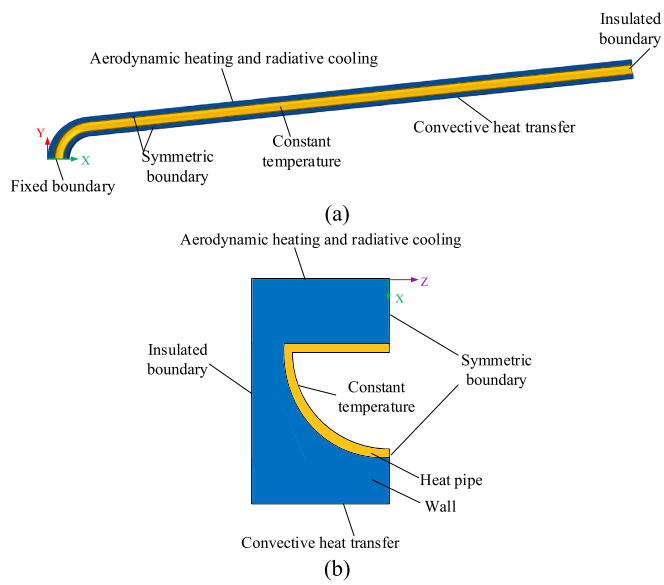

Fig. 2. Schematic diagram of the reusable heat-pipe cooled thermal protection system: (a) front view; (b) cross section.

The initial temperature of the reusable heat-pipe cooled thermal protection system is given by:

$$T(x,y,z,t)|_{t=0} = T_0(x,y,z) \tag{2}$$

The steam chamber of the reusable heat-pipe cooled thermal protection system has a very high isothermal performance, after the heat pipe enters the working state. Hence, suppose the inner wall of the heat pipe is isothermal. The aerodynamic heating and radiation boundary conditions are imposed on the upper surface of the reusable heat-pipe cooled thermal protection system, and the lower surface is subjected to the convective heat transfer boundary condition. The other boundaries are adiabatic.

$$T_i = \overline{T} \tag{3}$$

$$-k(T)\frac{\partial T}{\partial x} - k(T)\frac{\partial T}{\partial y} - k(T)\frac{\partial T}{\partial z} = q \tag{4}$$

$$-k(T)\frac{\partial T}{\partial x} - k(T)\frac{\partial T}{\partial y} - k(T)\frac{\partial T}{\partial z} = h(\mathrm{T}_l - \mathrm{T}_\infty) \tag{5}$$

The radiative heat flux imposed on the upper surface of the reusable heat-pipe cooled thermal protection system can be expressed as:

$$q_r = \varepsilon_r \sigma_b \left(T_r^4 - T_\infty^4\right) \tag{6}$$

In Eqs. (1)–(6), k, ρ, and c_p are the thermal conductivity, density, and specific heat capacity, respectively. T represents the temperature, Q denotes the volumetric heat source, and t is the time. x, y, and z are the spatial coordinates. q and h stand for the heat flux and the convective heat transfer coefficient, respectively. ε_r denotes the emissivity, and σ_b is the Stefan-Boltzmann constant, 5.67×10^{-8} W/(m^2·K^4). Subscript ∞ is the ambient.

The equilibrium equations of stress field analysis are:

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + f_x = 0 \tag{7}$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} + f_y = 0 \tag{8}$$

$$\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} + f_z = 0 \tag{9}$$

The initial strain caused by thermal deformation can be written as:

$$\boldsymbol{\varepsilon}_0 = \alpha(T - T_0)[1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0]^{\mathrm{T}} \tag{10}$$

The stress-strain relationship with initial strain can be expressed as follows:

$$\boldsymbol{\sigma} = \boldsymbol{D}(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_0) \tag{11}$$

The prescribed mechanical boundary conditions of the reusable heat-pipe cooled thermal protection system are fixed boundary condition and symmetric boundary condition, which are as shown in Fig. 2.

$$u(x,0,z) = v(x,0,z) = w(x,0,z) = 0 \tag{12}$$

$$w(x,y,0) = 0, \frac{\partial u}{\partial z} = \frac{\partial v}{\partial z} = 0 \tag{13}$$

In Eqs. (7)–(13), f denotes body force, σ and τ are normal stress and shear stress, respectively. ε stands for strain. α represents thermal expansion coefficient, and $\boldsymbol{D}$ is the constitutive matrix. u, v, and w are the displacement components along the x, y, and z directions, respectively.

The thermal contact resistance between the wall and heat pipe of the reusable heat-pipe cooled thermal protection system is very sensitive to interface pressure, temperature, and surface roughness [30]. Therefore, we assume the thermal contact resistance with a functional form as the following equation. The variation of spatial position represents surface roughness properties.

$$R = 1 \times 10^{-4} \exp(aP)\left(7 \times 10^{-3}T\right)^b \sin(cx)\left[-\ln\left(10y + 10^{-6}\right)\right]\cos^2(dz) + g \tag{14}$$

In Eq. (14), R represents the thermal contact resistance, and P is the interface pressure. The constant coefficients a, b, c, d, and g are −0.738, −0.427, 786, 2.54×10^5, and 6.92×10^{-5}, respectively.

2.2. Inverse analysis method

The five constant coefficients a, b, c, d, and g in Eq. (14) of the thermal contact resistance for the reusable heat-pipe cooled thermal protection system are simultaneously recovered based on the boundary temperature measurements by using the gradient-based algorithm [31] in this work. The dimensionless objective function can be defined by the following equation.

$$F(\boldsymbol{X}) = \sqrt{\frac{1}{M}\sum_{i=1}^{M}\left(\frac{T_i(\boldsymbol{X}) - T_i^*}{T_i^*}\right)^2}, \boldsymbol{X} = (a,b,c,d,g) \tag{15}$$

where M is the number of measurements, T_i is the calculated temperature, and T^* i stands for the measured temperature.

The inverted parameters can be updated by Eq. (16).

$$\boldsymbol{X}^{p+1} = \boldsymbol{X}^p + \omega\boldsymbol{\theta} \tag{16}$$

where

$$\boldsymbol{\theta} = \left[\boldsymbol{J}^{\mathrm{T}}\boldsymbol{J} + \mu\, diag\left(\boldsymbol{J}^{\mathrm{T}}\boldsymbol{J}\right)\right]^{-1}\boldsymbol{J}^{\mathrm{T}}[\boldsymbol{T}^* - \boldsymbol{T}(\boldsymbol{X})] \tag{17}$$

in which superscript p is the iteration number, and ω is the relaxation factor. $\boldsymbol{J}$ is sensitivity coefficients matrix, i.e., the derivative of the measurements with respect to the identified parameters. *diag* stands for diagonal elements, and μ stands for damping factor [32].

$$\boldsymbol{J}=\begin{bmatrix}\frac{\partial T_1}{\partial a} & \frac{\partial T_1}{\partial b} & \cdots & \frac{\partial T_1}{\partial g}\\ \frac{\partial T_2}{\partial a} & \frac{\partial T_2}{\partial b} & \cdots & \frac{\partial T_2}{\partial g}\\ \vdots & \vdots & \ddots & \vdots\\ \frac{\partial T_M}{\partial a} & \frac{\partial T_M}{\partial b} & \cdots & \frac{\partial T_M}{\partial g}\end{bmatrix} \tag{18}$$

The iteration of the inverse thermo-mechanical coupling problem is terminated until the following inequality holds.

$$F<\zeta \text{ or } \left|F^{p+1}-F^{p}\right|<\zeta \tag{19}$$

in which ζ is a small positive number, and in this work is 10^{-8}.

Temperature distribution and stress field influence each other. Therefore, the fully coupled thermal-stress analysis for the 3D reusable heat-pipe cooled thermal protection system is performed in the commercial finite element software ABAQUS. The calculation process of the inverse thermo-mechanical coupling problem is as follows.

(1) Set initial guess values of five inverted parameters, and give boundary measurement temperatures.
(2) Perform the thermo-mechanical coupling analysis by ABAQUS to obtain the calculated temperature values.
(3) Calculate the dimensionless objective function according to Eq. (15).
(4) Check the convergence criterion in Eq. (19). Stop the iteration if the inequality is satisfied. Otherwise, move to Step 5.
(5) Determine the variable damping factor, and calculate Eq. (17) to obtain updated vector $\boldsymbol{\theta}$ of recovered parameters.
(6) Update five recovered parameters according to Eq. (16). Then, go back to Step 2.

3. Validation of the present methodology with available experimental data

The proposed method is first validated by using available one-dimensional (1D) experimental data in Ref. [14], before its application in predicting thermal contact resistance in the 3D reusable heat-pipe cooled thermal protection system. A 1D steady state heat conduction experiment was performed to predict the constant thermal contact resistance for Ti–6Al–4V alloy in Ref. [14]. The physical model of the test specimens is shown in Fig. 3, and the temperature-dependent thermal conductivities of the experiment material are listed in Table 1. Three cases under different heating temperatures as listed in Table 2 are used to validate the present approach.

In the inversion, T_1 and T_6 are taken as boundary conditions, and T_3 and T_4 are taken as measurements. The constant thermal contact resistance is identified by solving the one-dimensional steady state inverse heat conduction problem. Note that the rough surfaces do not need to be reconstructed in estimating thermal contact resistance using the inversion method.

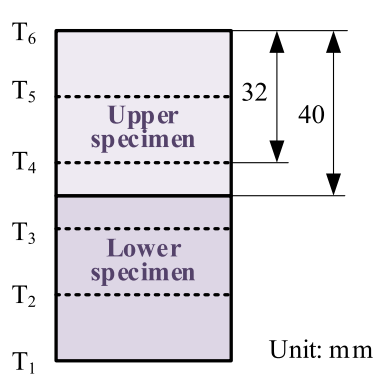

Fig. 3. Dimensions of experiment specimens [14].

Table 1
Thermal conductivity of Ti–6Al–4V alloy with temperatures [14].

Temperature (K)	Thermal conductivity (W/(m·K))
293.15	6.8
373.15	7.4
473.15	8.7
573.15	9.8
673.15	10.3
773.15	11.8

Table 2
The experimental measurements under different heating temperatures [14].

Case numbers	T_1 (K)	T_3 (K)	T_4 (K)	T_6 (K)
A	598.05	517.05	456.95	364.25
B	679.15	580.35	504.95	386.25
C	761.25	646.15	555.25	410.65

The results calculated by the present methodology are compared with the experimental measured TCR in Ref. [14] as listed in Table 3. It can be seen from Table 3 that the prediction results agree well with the experimental data among the three cases, and the maximum deviation is 2.49 %, indicating that the proposed method has high accuracy in determining thermal contact resistance at the interfaces.

4. Thermo-mechanical coupling analysis with thermal contact resistance

Then, the transient thermo-mechanical coupling analysis with thermal contact resistance of a 3D reusable heat-pipe cooled thermal protection system is described in the section. The boundary conditions of the thermal protection system are illustrated in Figs. 2, and Fig. 4 shows the contour parameters. All physical parameters and conditions are known for the direct problem. The inner wall of the heat pipe is specified with a temperature of 1000 K. Fig. 5 demonstrates the approximate aerodynamic heating condition on the upper surface. The emissivity in the radiative heat dissipation boundary condition is 0.8, and the ambient temperature is 300 K. The convective heat transfer coefficient on the lower surface is 20 W/(m^2·K). The temperature of the reusable thermal protection system is initialized to 300 K, and the total computation time is 5 s. Material properties of the reusable heat-pipe cooled thermal protection system are provided in Table 4.

Temperatures at five points and von Mises stresses along Path 1 as marked in Fig. 4 are selected to perform the grid and time step independence verification, and the calculation results are shown in Figs. 6 and 7.

It can be seen from Figs. 6 and 7 that the calculated results are stable with grid size and time step, which demonstrates that the mesh size and time step have slight effects on the computation results. 8880 elements and 0.05 s time step are applied for the following inversion computations in consideration of precision and efficiency.

The distributions of temperature and von Mises stress of the 3D reusable heat-pipe cooled thermal protection system are shown in Figs. 8

Table 3
Comparison of the identification results and the experimental data.

Case numbers	TCR × 10^4 (m^2 K/W)		Deviation (%)
	Experiment [14]	Inversion	
A	7.75	7.595	2.00
B	7.30	7.445	1.99
C	7.15	6.972	2.49

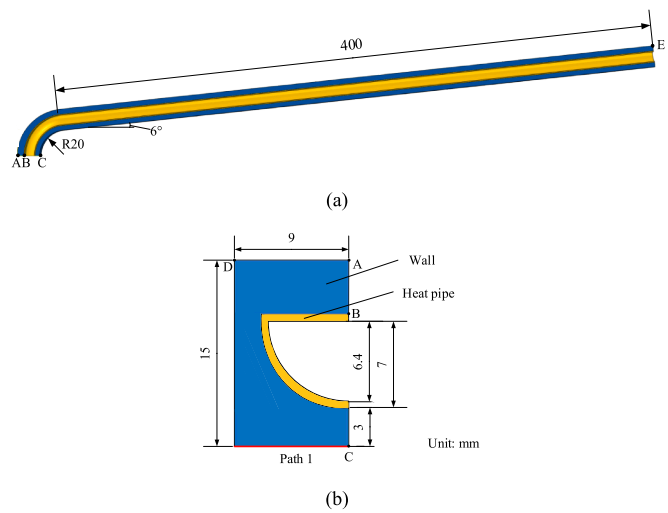

Fig. 4. Size parameters of the reusable thermal protection system.

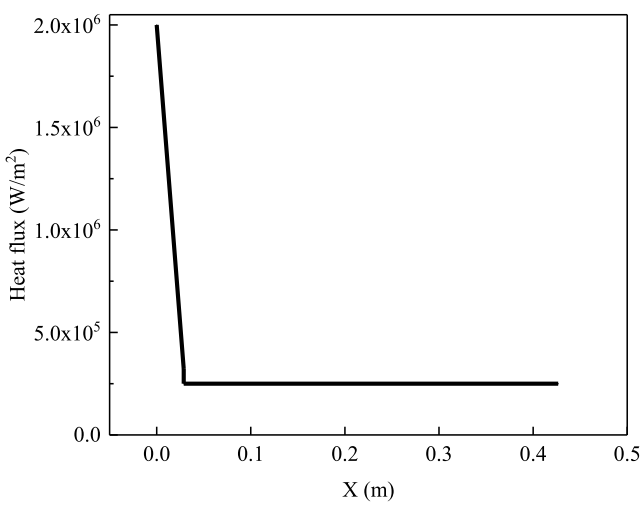

Fig. 5. Description of the aerodynamic heating condition.

Table 4
Material properties of reusable heat-pipe cooled thermal protection system.

Structures	Wall	Heat pipe
Thermal conductivity (W/(m·K))	67.7	13.9
Specific heat capacity (J/(kg·K))	1200	500
Density (kg/m^3)	1800	8430
Elastic modular (Pa)	3.3×10^{10}	2.01×10^{11}
Poisson ratio	0.3	0.27
Thermal expansion coefficient (1/K)	5.1×10^{-7}	1.2×10^{-6}

and 9. It can be observed from Fig. 8 that the temperature of the reusable heat-pipe cooled thermal protection system increases with computation time, due to the effect of the aerodynamic heating condition. The maximum von Mises stress is found in the stagnation area of the reusable heat-pipe cooled thermal protection system as shown in Fig. 9. This is attributed to the comprehensive influence of the temperature field and the fixed boundary condition.

5. Identification of TCR in the reusable heat-pipe cooled thermal protection system

The thermal contact resistance with a functional form in the reusable heat-pipe cooled thermal protection system is identified, and the good performance of the present method is demonstrated in this section.

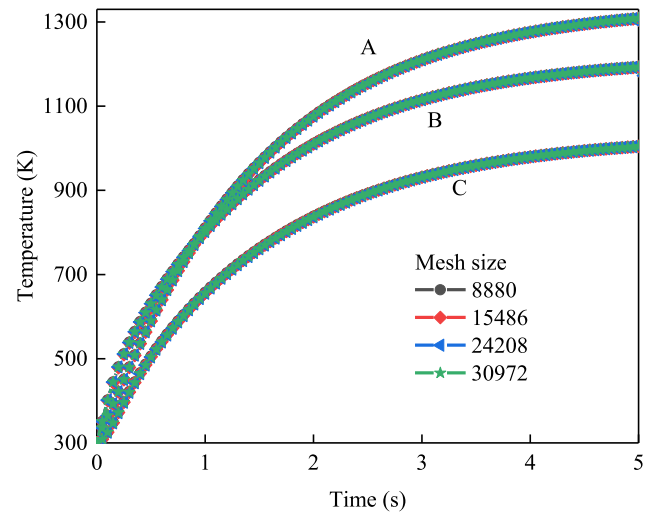

Fig. 6. Grid independence verification of heat-pipe cooled thermal protection system.

In the inversion problem, the thermal contact resistance is estimated based on the measurement history captured by the thermal sensors. To avoid damage to the specimen and facilitate operation, the positions of thermal sensor are only set on the lower surface of the reusable heat-pipe cooled thermal protection system in the work. The interval between temperature measurements is 0.5 s.

The effect of thermal sensor numbers on estimation is studied to obtain satisfactory results. Table 5 lists the identification results, and the initial guess values of a, b, c, d, and g are -0.7, -0.4, 700, 2.5×10^5, and 7×10^{-5}, respectively. Fig. 10 shows the sensitivity coefficients at the first iteration, when the number of thermal sensors is 15.

As can be observed in Fig. 10, the sensitivity coefficients are relatively high, which provides a guarantee for accurate identification. It can be seen from Table 5 that the iteration process converges, and the precision of the inversion is high, if the number of measurements is more than that of recovered parameters. Additionally, the estimated accuracy does not obviously increase with the increase of the number of thermal sensors. In the following numerical simulations, 15 thermal sensors are taken and there are 150 measurements.

Then, the results of the present method and the least square method are compared. The initial guess values of a, b, c, d, and g are -0.8, -0.5, 900, 2.5×10^5, and 9×10^{-5}, respectively. The relaxation factor is 1.0. The identification is implemented on a computer with CPU AMD Ryzen 7 5800H, 3.20 GHz, 16.0 GB of RAM. The thermal contact resistance identification results are listed in Table 6. It can be seen that the iteration number and computational time of the present method are small, which indicates that the method has high efficiency in solving the inverse thermo-mechanical coupling problem.

The gradient-based algorithm requires to set initial guess values to start the iterations, and it has been found that the solution of inverse problems using the method is very dependent on initial guess values [33]. Therefore, the effect of the initial guess values on the estimation results is investigated as listed in Table 7.

As can be seen from Table 7, satisfactory convergence results could be obtained, when initial guess values are selected as in groups 1 to 3. The iterative process diverges, when initial guess values are as shown in group 4. It can be concluded that appropriate initial guess values are necessary for the inverse thermo-mechanical coupling analysis, because the main limitation of the inversion method is that it may converge to a local minimum [34]. In general, the proposed method has good convergence stability for the estimation of thermal contact resistance by

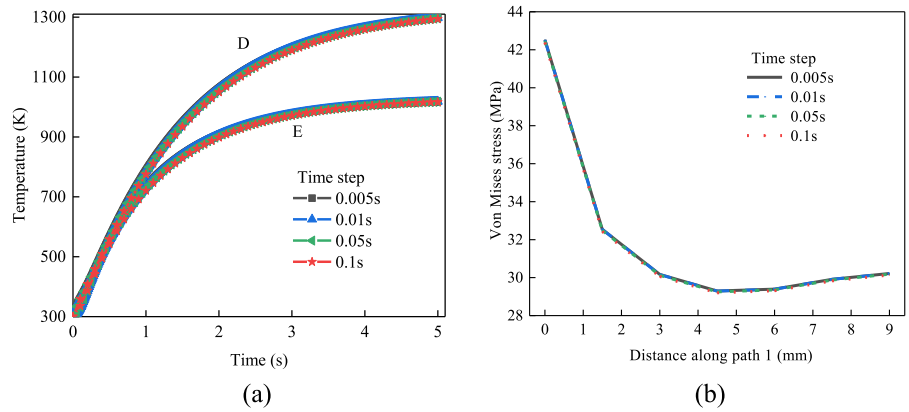

Fig. 7. Time step independence verification: (a) temperatures at different time; (b) von Mises stresses at 5s.

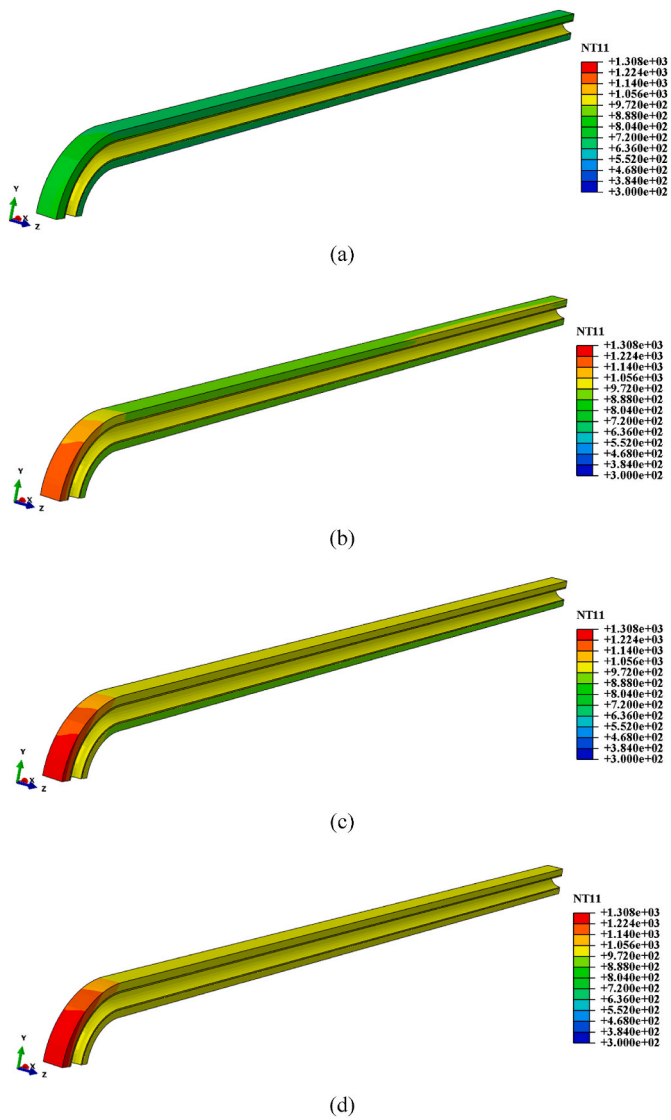

Fig. 8. Temperature distributions of heat-pipe cooled thermal protection system at different time: (a) 1 s; (b) 3 s; (c) 4 s; (d) 5 s.

solving inverse thermo-mechanical coupling problems.

The convergence process of the recovered parameters is shown in Fig. 11, when the initial guess value is group 2 marked in Table 7. It can be observed that five inverted parameters converge to the truth values after 13 iterations, and the final objective function is 7.43×10^{-9}, which indicates the excellent accuracy of the proposed method.

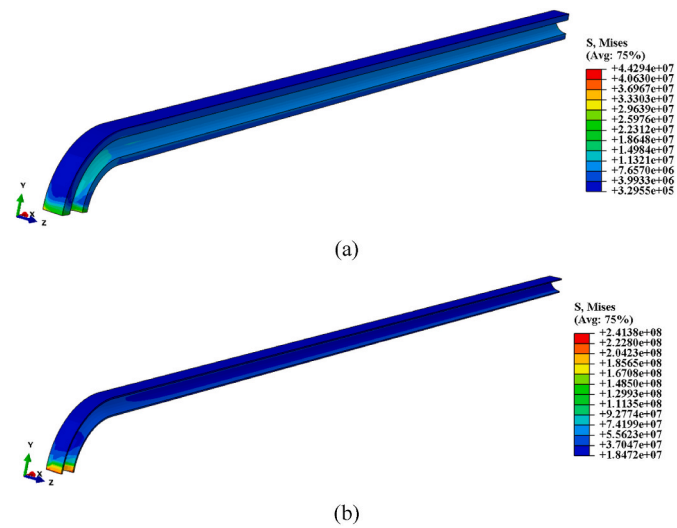

Fig. 9. Von Mises stress distributions of heat-pipe cooled thermal protection system at 5 s: (a) wall; (b) heat pipe.

Table 5
Identification results with different thermal sensor numbers.

Thermal sensor numbers	Convergence	Iteration numbers	Final F
5	Yes	7	0
10	Yes	6	0
15	Yes	6	1.26×10^{-8}
20	Yes	7	1.35×10^{-8}

Actual temperature measurement process is accompanied by measurement errors. Hence, in order to evaluate the influence of measurement error on the present method, random test errors [35,36] are imposed in the exact measurement temperatures.

$$T = T^{*}\left(1 + \frac{\xi\eta}{2.576}\right) \tag{20}$$

in which ξ is the measurement error, and η is a normal distributed random variable with zero mean and constant standard deviation. 2.576 results from the fact that 99 % of a normally distributed population is contained within ±2.576 standard deviation of the mean [37].

The estimation results with different measurement errors are listed in

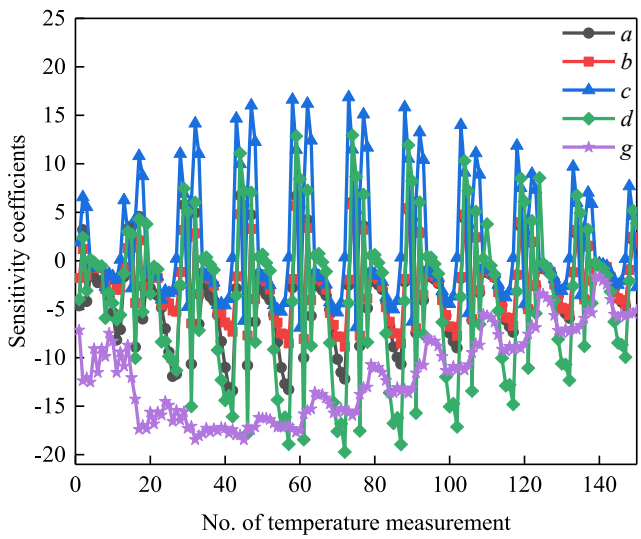

Fig. 10. Sensitivity coefficients when the number of thermal sensors is 15.

Table 6
Identification results of different prediction methods.

Methods	Iteration number	Computational time, s
Least square method	10	43070
Present method	7	34237

Table 7
Estimation results with different initial guess values.

Groups	Initial guess values					Convergence	Final F
	a	b	c	$d \times 10^{-5}$	$g \times 10^{5}$		
1	−1.2	−0.7	1000	4.0	12	Yes	2.34×10^{-8}
2	−0.9	−0.5	900	3.0	9	Yes	7.43×10^{-9}
3	−0.7	−0.4	650	2.4	6	Yes	7.43×10^{-9}
4	−0.4	−0.4	430	0.1	0.1	No	–

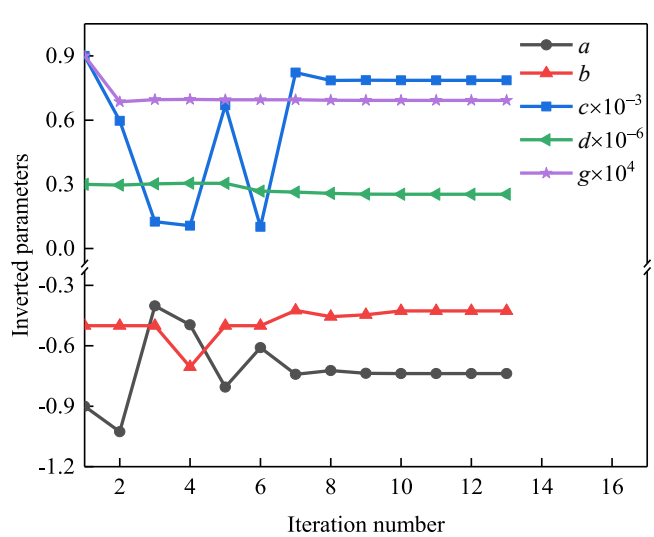

Fig. 11. Convergence process of the recovered parameters.

Table 8
Identification results with different measurement errors.

Inverted parameters	ξ=0.5 %	ξ=1.0 %	ξ=1.5 %
a	−0.717	−0.700	−0.686
b	−0.436	−0.424	−0.411
c	786.474	786.204	785.714
d	2.570×10^{5}	2.590×10^{5}	2.602×10^{5}
g	6.925×10^{-5}	6.929×10^{-5}	6.932×10^{-5}

Table 8. A standard deviation of 0.5 K is considered when generating the measurement data, and Fig. 12 illustrates the normal distributed random variable η when measurement error is 1.0 %.

It is not surprising that a higher measurement error results in a lower precision for the estimated results as shown in Table 8. However, the identification results are still within the acceptable range, when measurement error is 1.5 %. It can be concluded that the inversion method remains accurate for determining the thermal contact resistance of the reusable heat-pipe cooled thermal protection system, although measurement errors deteriorate its performance.

6. Conclusions

This work proposed an inverse thermo-mechanical coupling method to identify the thermal contact resistance (TCR) of a three-dimensional (3D) reusable heat-pipe cooled thermal protection system, based on boundary measurement points. The thermal contact resistance with a functional form to be estimated depends on contact pressure, interface temperature, and spatial position. The available experimental data is employed to validate the effectiveness and accuracy of the inversion method, and the deviation between experimental data and identified thermal contact resistance is less than 2.5 %. The present method shows excellent estimation precision if the number of measurements is more than that of identification parameters for the reusable heat-pipe cooled thermal protection system. Compared with the least square method, the present method has high efficiency in predicting the thermal contact resistance with a functional form. The effects of initial guess values and measurement errors are numerically investigated by solving transient inverse thermo-mechanical coupling problems, and the results show that the present method has good convergence stability and robustness in determining the TCR of the 3D reusable heat-pipe cooled thermal protection system.

The present work provides an effective method for the estimation of

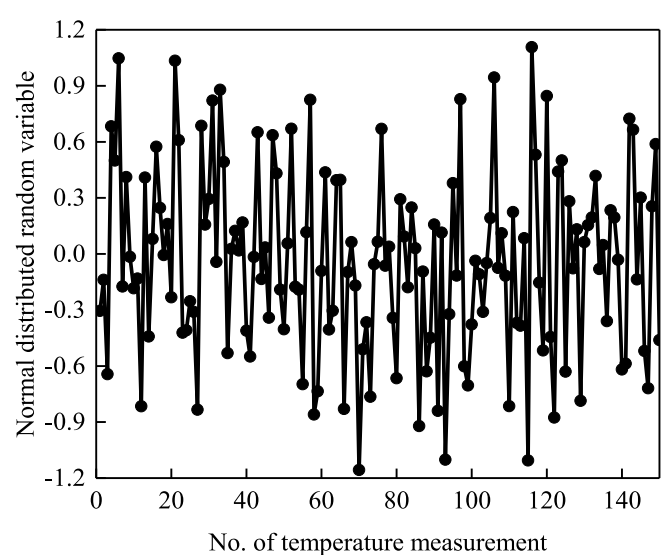

Fig. 12. Normal distributed random variable when measurement error is 1.0 %.

the thermal contact resistance in thermal management systems, which also offers important references for the optimization design of the 3D reusable heat-pipe cooled thermal protection system used in various operation environments in aerospace engineering. In the future, the present method can be used to predict thermal contact resistance taking into account factors such as material properties.

CRediT authorship contribution statement

Chunyun Zhang: Writing – original draft. **Peng Yu:** Data curation. **Chengbao Sun:** Software. **Haifeng Peng:** Data curation. **Miao Cui:** Supervision, Writing – review & editing. **Bingbing Xu:** Data curation, Formal analysis, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] M. Seong, I. Hwang, S. Park, H. Jang, G. Choi, J. Kim, S.K. Kim, G.H. Kim, J. Yeo, H.E. Jeong, Enhanced thermal transport across self-interfacing van der Waals contacts in flexible thermal devices, Adv. Funct. Mater. 31 (48) (2021) 2107023.
[2] P. Gimenez, A. Jove, C. Prieto, S. Fereres, Effect of an increased thermal contact resistance in a salt PCM-graphite foam composite TES system, Renew. Energy 106 (2017) 321–334.
[3] L.X. Gu, Y.F. Wang, S.B. Shi, C.X. Dai, An approximate analytical method for nonlinear transient heat transfer through a metallic thermal protection system, Int. J. Heat Mass Transf. 103 (2016) 582–593.
[4] D.H. Liu, Y. Luo, X.C. Shang, Experimental investigation of high temperature thermal contact resistance between high thermal conductivity C/C material and Inconel 600, Int. J. Heat Mass Transf. 80 (2015) 407–410.
[5] J. Guo, X.N. Chen, Z.G. Qu, Q.L. Ren, Reverse identification method for simultaneous estimation of thermal conductivity and thermal contact conductance of multilayered composites, Int. J. Heat Mass Transf. 173 (2021) 121244.
[6] H.P. Liu, W.Q. Liu, Thermal–structural analysis of the platelet heat-pipe-cooled leading edge of hypersonic vehicle, Acta Astronaut. 127 (2016) 13–19.
[7] F.Z. Sun, P. Zhang, H. Wang, J.R. Shen, L.Q. Nie, Experimental study of thermal contact resistance between aluminium alloy and ADP crystal under vacuum environment, Appl. Therm. Eng. 155 (2019) 563–574.
[8] D.X. Wu, X.S. Zhang, Study on the methods of tuning thermal contact resistance of nanoparticles, Physica B 627 (2022) 413551.
[9] A. Aalilija, C.-A. Gandin, E. Hachem, A simple and efficient numerical model for thermal contact resistance based on diffuse interface immersed boundary method, Int. J. Therm. Sci. 166 (2021) 106817.
[10] M.J. Chen, Q. Li, P. Zhang, Experimental investigation of high temperature thermal contact resistance of thin disk samples using infrared camera in vacuum condition, Int. J. Heat Mass Transf. 157 (2020) 119749.
[11] Y.Q. Xian, P. Zhang, S.P. Zhai, P. Yuan, D.G. Yang, Experimental characterization methods for thermal contact resistance: a review, Appl. Therm. Eng. 130 (2018) 1530–1548.
[12] K.F. Xie, Y. Cui, X.N. Ren, Y.X. Chen, J. Cai, W.B. Hu, Flash DSC characterization of thermal contact resistance and cross-plane thermal conductivity of micrometer-thin films, Thermochim. Acta 724 (2023) 179493.
[13] P.G. Siddappa, A. Tariq, Contact area and thermal conductance estimation based on the actual surface roughness measurement, Tribol. Int. 148 (2020) 106358.
[14] Y.J. Dai, J.J. Gou, X.J. Ren, F. Bai, W.Z. Fang, W.Q. Tao, A test-validated prediction model of thermal contact resistance for Ti-6Al-4V alloy, Appl. Energy 228 (2018) 1601–1617.
[15] J.N. Zhao, D. Wei, A.Q. Gao, H.L. Dong, Y.B. Bao, Y.M. Jiang, D.H. Liu, Thermal rectification enhancement of bi-segment thermal rectifier based on stress induced interface thermal contact resistance, Appl. Therm. Eng. 176 (2020) 115410.
[16] R.C. Dhuley, Pressed copper and gold-plated copper contacts at low temperatures – a review of thermal contact resistance, Cryogenics 101 (2019) 111–124.
[17] S. Wen, H. Qi, Z.T. Niu, Y.T. Ren, L.M. Ruan, Fast non-intrusive estimation of liquid-phase interface and boundary heat flux in participating media by the Kalman filtering methods, Int. J. Heat Mass Transf. 142 (2019) 118418.
[18] B.A. Tourn, J.C. Álvarez Hostos, V.D. Fachinotti, A modified sequential gradient-based method for the inverse estimation of transient heat transfer coefficients in non-linear one-dimensional heat conduction problems, Int. Commun. Heat Mass Transf. 127 (2021) 105488.
[19] B. Yu, C. Xu, W.A. Yao, Z. Meng, Estimation of boundary condition on the furnace inner wall based on precise integration BEM without iteration, Int. J. Heat Mass Transf. 122 (2018) 823–845.
[20] Z. Kaczor, Z. Bulinski, S. Sobek, S. Werle, Application of inverse methodology to estimate unknown parameters of the mathematical model of biomass solar pyrolysis, Renew. Energy 163 (2021) 858–869.
[21] N. BniLam, R. Al-Khoury, Parameter identification algorithm for ground source heat pump systems, Appl. Energy 264 (2020) 114712.
[22] M. Parikh, S. Shah, H. Vaghela, A.K. Parwani, A comprehensive experimental and numerical estimation of thermal contact conductance, Int. J. Therm. Sci. 172 (2022) 107285.
[23] K. Goudarzi, A. Moosaei, M. Gharaati, Applying artificial neural networks (ANN) to the estimation of thermal contact conductance in the exhaust valve of internal combustion engine, Appl. Therm. Eng. 87 (2015) 688–697.
[24] W. Kanjanakijkasem, Estimation of spatially varying thermal contact resistance from finite element solutions of boundary inverse heat conduction problems split along material interface, Appl. Therm. Eng. 106 (2016) 731–742.
[25] C. Fieberg, R. Kneer, Determination of thermal contact resistance from transient temperature measurements, Int. J. Heat Mass Transf. 51 (5–6) (2008) 1017–1023.
[26] Z.W. Liu, G. Wang, J. Yi, Study on heat transfer behaviors between Al-Mg-Si alloy and die material at different contact conditions based on inverse heat conduction algorithm, J. Mater. Res. Technol. 9 (2) (2020) 1918–1928.
[27] R. Brociek, E. Hetmaniok, D. Slota, Reconstruction of aerothermal heating for the thermal protection system of a reusable launch vehicle, Appl. Therm. Eng. 219 (2023) 119405.
[28] B.W. Zhang, J. Mei, C.Y. Zhang, M. Cui, X.W. Gao, Y.W. Zhang, A general method for predicting the bank thickness of a smelting furnace with phase change, Appl. Therm. Eng. 162 (2019) 114219.
[29] Z.W. Fu, B. Han, Y. Chen, Levenberg–Marquardt method with general convex penalty for nonlinear inverse problems, J. Comput. Appl. Math. 404 (2022) 113771.
[30] D.H. Liu, X.C. Shang, Effect of thermal contact resistance on the performance of heat-pipe-cooled thermal protection structures, Acta Aeronautica Astronautica Sinica 33 (10) (2012) 1834–1841 (in Chinese).
[31] M. Cui, K. Yang, X.L. Xu, S.D. Wang, X.W. Gao, A modified Levenberg–Marquardt algorithm for simultaneous estimation of multi-parameters of boundary heat flux by solving transient nonlinear inverse heat conduction problems, Int. J. Heat Mass Transf. 97 (2016) 908–916.
[32] M. Cui, Y. Zhao, B.B. Xu, X.W. Gao, A new approach for determining damping factors in Levenberg-Marquardt algorithm for solving an inverse heat conduction problem, Int. J. Heat Mass Transf. 107 (2017) 747–754.
[33] R. Sajedi, J. Faraji, F. Kowsary, A new damping strategy of Levenberg-Marquardt algorithm with a fuzzy method for inverse heat transfer problem parameter estimation, Int. Commun. Heat Mass Transf. 126 (2021) 105433.
[34] C. Kanzow, N. Yamashita, M. Fukushima, Levenberg–Marquardt methods with strong local convergence properties for solving nonlinear equations with convex constraints, J. Comput. Appl. Math. 172 (2) (2004) 375–397.
[35] S.B. Wan, K. Wang, P. Xu, Y.J. Huang, Numerical and experimental verification of the single neural adaptive PID real-time inverse method for solving inverse heat conduction problems, Int. J. Heat Mass Transf. 189 (2022) 122657.
[36] C.Y. Zhang, J. Mei, Y.S. Bai, M. Cui, H.F. Peng, X.W. Gao, Simultaneous identification of multi-parameter for power hardening elastoplastic problems in three-dimensional geometries, Eng. Comput. 39 (8) (2022) 2990–3011.
[37] M. Cui, J. Mei, B.W. Zhang, B.B. Xu, L. Zhou, Y.W. Zhang, Inverse identification of boundary conditions in a scramjet combustor with a regenerative cooling system, Appl. Therm. Eng. 134 (2018) 555–563.